

COMPACT SECONDARY SUBSTATION

PCSS

Prefabricated MV/LV substation — switchgear, transformer and LV board in one kiosk

Up to 24 kV · up to 2,500 kVA



2,500 kVA

TRANSFORMER
RATINGS

24 kV

MV RATED VOLTAGE

5,000 A

LV SWITCHBOARD
RATING

IEC

62271-202 DESIGN

01 · COMPANY PROFILE

ENGINEERING THE FUTURE OF POWER WITH PRECISION AND INNOVATION

Since our inception in 2012, Powerline has been driven by a singular mission: to serve as the reliable backbone of the regional and Egyptian power sector. We are not merely a contracting and supply firm — we are a fully integrated engineering partner specialising in the design and manufacture of Medium Voltage (MV) and Low Voltage (LV) power distribution systems, adhering to the most stringent international technical standards.

A Legacy of Reliability

Built on foundations of commitment and precision, Powerline has established itself as a hallmark of quality in major national and industrial projects. Our track record reflects a consistent ability to overcome complex engineering challenges and deliver customised solutions for the modern electrical grid.

Solutions Beyond Expectations

Our MV switchgear, Ring Main Units and LV main and sub-main distribution boards are characterised by exceptional operational efficiency and a long life cycle. We bridge mechanical durability with digital intelligence, ready for integration with the latest smart-grid and control technologies.

Quality is not a department — it is our work culture



Production & Manufacturing

Advanced production lines ensure the highest accuracy in the assembly and calibration of MV & LV switchgear, in full compliance with IEC international standards.



Engineering & Design

Our design engineers develop layouts that prioritise maximum safety, operational continuity and seamless serviceability across the installation's life.



Testing & Quality Control

Every unit leaving our facilities undergoes rigorous routine testing to ensure durability under the harshest operating conditions — total peace of mind for our clients' investments.

2012

ESTABLISHED · CAIRO

2MANUFACTURING
FACILITIES**MV+LV**

INTEGRATED PORTFOLIO

IECCERTIFIED QUALITY
SYSTEM

" Powerline: Performance Excellence, Execution Precision & Sustainable Solutions. "

TABLE OF CONTENTS

01	About Powerline	02	10	MV Compartment	13
02	PCSS at a Glance	04	11	Transformer Compartment	14
03	Field of Application	05	12	LV Compartment	15
04	One Substation, Three Compartments	06	13	Smart Grid Solutions	16
05	Safety by Design	07	14	Installation	17
06	Design & Construction	08	15	Quality & Testing	18
07	Main Technical Characteristics	09	16	Configuration Guide	19
08	Standards	10	17	Contact & Ordering	20
09	Enclosure & Protection	11			

How to use this catalogue

Sections 06–09 cover the enclosure and its ratings. Sections 10–13 detail what goes inside each compartment. Sections 14–16 support planning, testing and ordering.

02 · PRODUCT OVERVIEW

PCSS AT A GLANCE

PCSS is a complete, factory-built MV/LV substation in one compact kiosk: medium voltage switchgear, distribution transformer and low voltage switchboard — delivered tested, ready to connect.

The compact secondary substation was created to help distribution utilities move from overhead to underground networks: the full functionality of a conventional substation, with a minimum of site work. In urban networks PCSS is internally operated for safe service; for rural networks, externally serviced arrangements minimise cost and footprint.

- Robust, lightweight galvanized-steel platform — easy transport over long distances, less lifting effort
- Compartmented design: MV switchgear, transformer and LV switchboard, each behind padlockable doors
- Effective natural ventilation and IP options for warm and cold climates
- Internal arc classified per the IEC range for CSS applications — safety for public and personnel
- Complete factory-delivered solution: only external connections on site
- SCADA / RTU connectivity — ready for the smart grid



PCSS compact secondary substation — Powerline factory

One responsibility, one test

Enclosure and equipment engineered, assembled and routine-tested together by Powerline — a single point of responsibility for the complete substation.

24 kV

MV RATED VOLTAGE

2,500 kVA

MAX. TRANSFORMER

5,000 A

LV RATED CURRENT

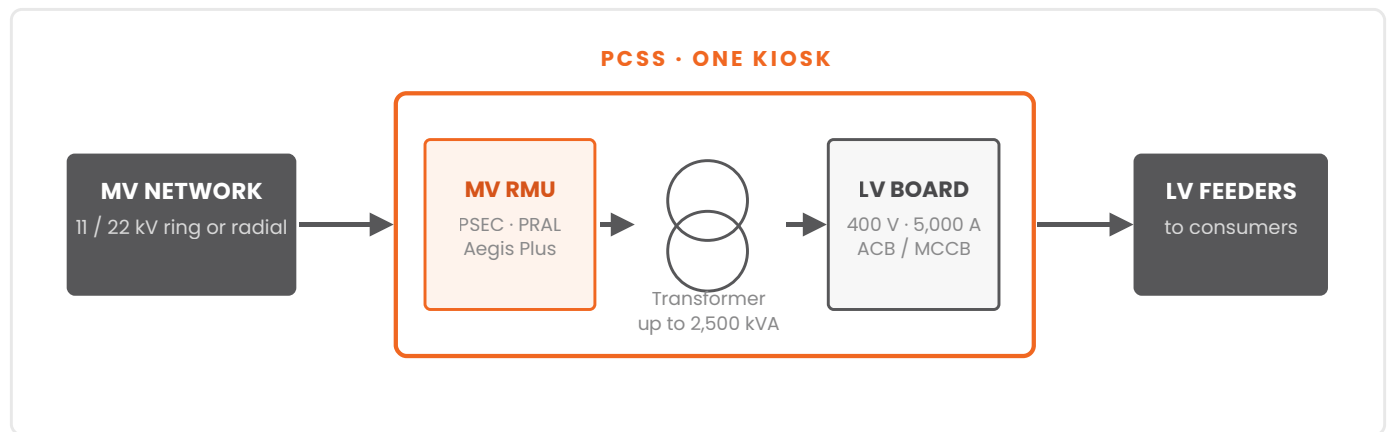
IP54MV / LV
COMPARTMENTS

Inside every PCSS: a Powerline RMU — **PSEC** (SF6 switch RMU), **PRAL** (air-insulated RMU) or **Aegis Plus** (GIS) — a distribution transformer up to 2,500 kVA, and a type-tested LV switchboard. See pages 13–15.

03 · APPLICATIONS

FIELD OF APPLICATION

PCSS receives medium voltage from the network, transforms it and distributes low voltage to consumers — the complete MV/LV substation, prefabricated. It is the natural building block wherever utilities move overhead lines underground, and wherever a safe, fast-to-deploy substation is needed.



Energy flow through the PCSS — MV switching, transformation and LV distribution in one enclosure

Typical duties

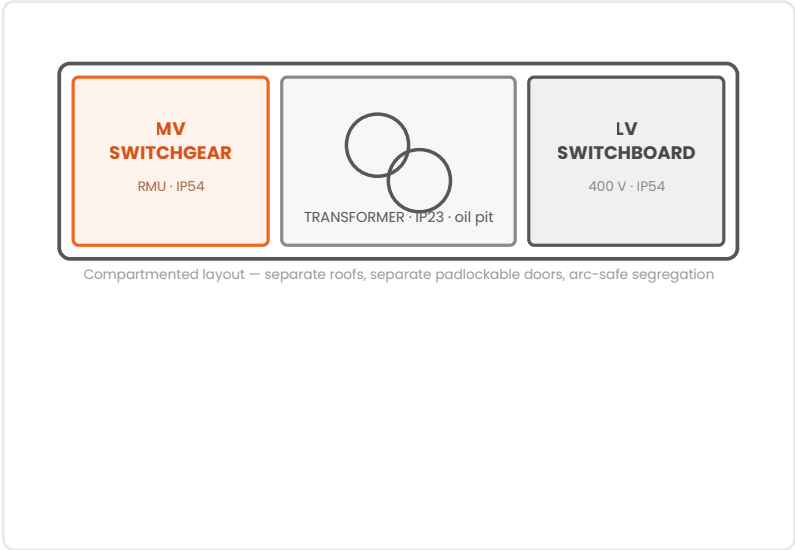
- MV/LV distribution nodes on urban underground networks — internally operated for safe servicing
- Rural electrification — externally serviced layouts minimise footprint and installation cost
- Temporary or fast-track power for construction and events
- Dedicated substations for compounds, facilities and industry

Typical environments

Utility distribution networks	Residential compounds
Commercial & mixed-use	Industrial plants
Solar & wind grid connection	Infrastructure projects
Rural & remote areas	
Coastal sites (stone-paint option)	

ONE SUBSTATION, THREE COMPARTMENTS

PCSS is divided into three segregated compartments — MV switchgear, distribution transformer and LV switchboard — complying with the latest IEC requirements for operator protection in the event of an internal arc. The transformer compartment determines the substation size and accepts oil or dry types.



Doors open — MV switchgear and transformer compartments

Compartment	Contents & features
MV switchgear	Ring main unit up to 24 kV — PSEC, PRAL or Aegis Plus; network switching and transformer protection ways; IP54
Transformer	Oil or dry type, up to 1,600 kVA standard / 2,500 kVA special design; smooth-air-flow cooling, movable roof section, oil collection pit; IP23
LV switchboard	Type-tested board up to 400 V / 5,000 A; incoming ACB, outgoing MCCBs or switch-fuses, metering and surge protection; IP54

3

SEGREGATED COMPARTMENTS

3

SEPARATE SLOPED ROOFS

100%

PADLOCKABLE DOORS

0

ACCESS TO LIVE PARTS

SAFETY BY DESIGN

The standard PCSS design has passed the internal arc classification test of the IEC range for CSS applications — the highest level of safety for the public walking past the kiosk and for the personnel operating it.



Internal arc classified

Type-tested internal arc behaviour of the enclosure protects both operators and the general public in the event of a fault inside.



Compartmented design

MV, transformer and LV zones are fully segregated; a fault or intervention in one compartment does not expose the others.



No unauthorised access

Robust pre-galvanized doors with locking systems on all compartments prevent unauthorised entry and vandalism.



No access to live parts

Every component is metal-enclosed and type-tested; operation activities are performed from outside the enclosure.



Earthing network

Internal earthing network rated up to 21 kA / 1 s bonds enclosure, equipment and doors per IEC 62271-202.



Environmental protection

Oil collection pit under the transformer traps insulating fluid in case of failure — protecting soil and groundwater.

IEC 62271-202

Every component inside — switchgear, transformer, LV board — is itself type-tested to its own product standard before the substation is verified as a whole. Safety is layered, not assumed.

DESIGN & CONSTRUCTION

An above-ground kiosk installed on a simple concrete base — roof, doors and ventilation engineered as one thermal and mechanical system.



Housing & material

- Galvanized sheet steel for high corrosion protection
- Stone-paint finish option for coastal areas
- Pre-treated, coated metal parts for outdoor service
- Robust steel structure — mechanically stable even under arc fault
- Lightweight platform: easy transport and lifting



Ventilation

- Natural ventilation — air intake through door openings
- Hot air extracted via louvers on the top sides
- Continuous circulation cools the distribution transformer
- Low enclosure temperature class — longer transformer life and higher usable rating
- IP class options for cold and warm climates



Roof & doors

- Separate roof over each compartment (MV / TX / LV)
- Sloped profile — no standing rain or liquids
- Movable transformer-roof section for easy installation and exchange
- Pre-galvanized sheet-steel doors, robust against unauthorised access
- Door design supports IP class and public arc protection

Why PCSS is a unique design

High safety for personnel, environment and equipment · isolated compartments with padlockable doors · complete factory-delivered solution with only external connections on site · SCADA / RTU-ready · increased corrosion resistance · routine-tested and type-tested · a portfolio covering all CSS applications.

1

CONCRETE PLINTH —
ALL CIVIL WORKS

100%

FACTORY ASSEMBLED
& TESTED

MAIN CHARACTERISTICS

Ratings of the standard PCSS layouts. Component-level data follows the installed equipment — see the PSEC, PRAL and Aegis Plus catalogues for MV switchgear details.

Characteristic	PCSS standard layouts
Maximum transformer rating	Up to 2,500 kVA
Rated voltage, MV	Up to 24 kV
Short-circuit withstand of internal earthing network	Up to 21 kA / 1 s
Transformer compartment protection degree	IP23
MV / LV compartment protection degree	IP54
Enclosure thermal class	10 K / 20 K
MV cabling	Cu conductors per EEHC specifications
MV terminations	Pre-moulded / cold-shrink / heat-shrink cable heads per customer request
LV connection to transformer	Busbar sized to the transformer rated power
Rated voltage of LV switchboard	Up to 400 V
Rated current of LV switchboard	Up to 5,000 A
Short-circuit withstand of LV busbar system	Up to 50 kA / 1 s

Urban & rural variants

Internally operated layouts for urban service teams; externally serviced layouts for minimum footprint in rural networks.

Climate ready

Ventilation design and IP options suit warm and cold climates; corrosion class allows installation across site conditions.

Remote ready

Connect PCSS to SCADA / RTU for remote monitoring and control — fewer site visits, maintenance-free routine.

STANDARDS & EVOLUTION

From IEC 61330 to IEC 62271-202

The first international standard for prefabricated HV/LV substations, IEC 61330, defined the ratings, design and operation rules that keep personnel safe and operations continuous — for cable-connected substations with transformers up to 1,600 kVA, including internal arc type testing.

Its successor, **IEC 62271-202 ed. 2**, governs today’s PCSS. Type tests are classified as mandatory, mandatory-where-applicable (per local regulation) and optional (agreed between manufacturer and user) — and every component installed inside must itself be type-tested to its own product standard first.

PCSS is designed and routine-tested to these definitions — giving utilities a substation with a verified, layered safety case rather than a collection of parts.

Standard	Scope
IEC 62271-202	High-voltage / low-voltage prefabricated substation
IEC 62271-1	Common specifications for HV switchgear
IEC 62271-200	AC metal-enclosed switchgear
IEC 62271-201	AC insulation-enclosed switchgear
IEC 60076	Power transformers
IEC 61439	Low voltage switchgear assemblies

Also influencing the design

IEC/TR 62271-208	Electromagnetic fields generated by a CSS
IEC 60529	Degrees of protection by enclosures (IP)
IEC 62262	Protection against external mechanical impact (IK)
IEC 60721	Classification of environmental conditions

ENCLOSURE & PROTECTION



Protection degree

MV and LV compartments are rated **IP54** — stable, controlled conditions for the switchgear and switchboard. The transformer compartment is rated **IP23** including doors and ventilation openings, balancing protection with cooling air flow.



Ventilation

Natural ventilation through intake louvers in doors or wall elements and extraction at the top — sufficient transformer cooling with no fans, no auxiliary power and nothing to maintain.



Door gaskets

For additional protection against dust being drawn into the MV and LV compartments, doors can be fitted with gaskets that raise the effective degree of protection.



Oil collection pit

Beneath the transformer, the oil pit traps insulating fluid in the event of a transformer failure — protecting the ambient environment and groundwater.

Enclosure thermal class

The enclosure class (**10 K / 20 K**) states how much the air around the transformer rises above ambient at rated load. Powerline's ventilation design targets the low class — cooler air means the transformer delivers more of its nameplate rating, ages slower and lives longer.

IP54

MV & LV
COMPARTMENTS

IP23

TRANSFORMER
COMPARTMENT

10K-20K

THERMAL CLASS
OPTIONS

SECTIONS 10 – 12

Inside PCSS

Three compartments, one engineered system. Powerline equips every PCSS with its own type-tested MV switchgear, a distribution transformer from approved manufacturers, and a type-tested LV switchboard — all per EEHC-approved material lists and the client's specification.

MV compartment · page 13

PSEC · PRAL · Aegis Plus ring main units, up to 24 kV

Transformer compartment · page 14

Oil or dry type, up to 2,500 kVA

LV compartment · page 15

Type-tested switchboard, up to 400 V · 5,000 A



MV SWITCHGEAR AND TRANSFORMER
COMPARTMENTS

24 kV

MV SWITCHGEAR

2,500

KVA MAX.
TRANSFORMER

5,000 A

LV SWITCHBOARD

EEHC

APPROVED MATERIALS

MV SWITCHGEAR

The MV compartment hosts a Powerline ring main unit up to 24 kV — providing network opening points for service and reconfiguration, and protecting the transformer per local rules and rating.

PSEC

SF6-switch RMU · 12 / 24 kV.

Compact air-insulated cubicle system with sealed-for-life SF6 three-position load break switches (ABB GSec or Murge MXGN). The workhorse choice for utility schemes.

PRAL

Air-insulated RMU · 12 / 24 kV.

Fully SF6-free switching built on the ABB NAL / NALF switch-disconnector — zero gas handling, simple end-of-life recycling.

AEGIS PLUS

GIS RMU · 12 / 24 kV.

Gas-insulated ring main unit for schemes requiring circuit breakers or load break switches in a sealed gas tank — minimal footprint, climate-immune.

Configured to the scheme

- Ring / network switching ways for open-ring and radial networks
- Transformer protection by switch-fuse combination — or circuit breaker for smaller ratings and adjustable protection
- Metering cubicles for utility revenue metering where required
- MV cabling in Cu per EEHC specifications; pre-moulded, cold-shrink or heat-shrink terminations per request

Full technical data in the dedicated **PSEC**, **PRAL** and **Aegis Plus** catalogues — the same units, the same options, factory-installed in your PCSS.



RMU installed in the PCSS MV compartment

DISTRIBUTION TRANSFORMER

The transformer compartment determines the substation size — and it is designed to fit all transformer types, oil or dry, from all approved manufacturers.

- **Oil-type or dry-type** distribution transformers, available on request
- **Up to 1,600 kVA as standard**; special designs up to **2,500 kVA** per Egyptian Electricity Holding Company standards
- Smooth air flow and natural cooling designed to meet the IEC temperature-rise requirements
- Movable roof section for easy installation, exchange and maintenance
- Oil collection pit underneath — environment and groundwater protected
- LV connection by busbar, sized to the transformer rated power
- PCSS can be lifted with the transformer installed

Approved manufacturers

Dimensioned to fit transformers from El Sewedy, Hitachi Energy, Egy-Trafo and other approved suppliers — per the EEHC-approved material lists and the client’s preference.

Characteristic	Specification
Type	Oil-immersed / dry-type
Standard ratings	Up to 1,600 kVA
Special design	Up to 2,500 kVA (per EEHC)
Product standard	IEC 60076
Compartment protection	IP23
Cooling	Natural ventilation (AN)
Oil containment	Collection pit under transformer

Cooler enclosure, stronger transformer

The low enclosure thermal class achieved by the ventilation design means the transformer can be worked closer to its full nameplate rating — more usable kVA from the same unit.

LV SWITCHBOARD

The LV switchboard distributes the transformed energy — type-tested to the latest IEC 61439, with the flexibility to serve anything from a simple feeder pillar duty to a fully metered, surge-protected distribution board.

- **Incoming switch** protects the transformer from LV network faults — and vice versa
- **Outgoing feeders** protected by switch-fuses or circuit breakers (ACB / MCCB)
- Feeder count and ratings configured to the transformer size and the customer’s needs
- Auxiliary equipment: measuring instruments, surge protection and any additional devices required
- Busbar system rated up to 5,000 A · 50 kA / 1 s

Component brands

ABB circuit breakers are recommended for main and outgoing positions, with full flexibility to switch to Hiemel or other brands per the consultant’s and client’s request — always within the EEHC-approved material lists.



LV switchboard — incoming ACB, outgoing feeders and controls

Characteristic	Specification
Rated voltage	Up to 400 V
Rated current	Up to 5,000 A
Busbar short-circuit withstand	Up to 50 kA / 1 s
Product standard	IEC 61439 (type-tested)
Compartment protection	IP54

SMART GRID SOLUTIONS

In line with the Egyptian Electricity Holding Company's drive towards a smart grid, PCSS is offered as an automated secondary substation — improving fault detection, isolation and restoration, voltage regulation, load balancing and asset protection.

L1

Monitor

Monitoring of the entire secondary substation, with current, voltage and energy measurement on the LV side.

L2

Monitor + Control

All Level 1 functions, plus remote control of the medium and low voltage primary apparatus.

L3

+ MV Measurement

All Level 2 functions, with accurate current, voltage and energy measurement on the MV side.

L4

+ Protection

The complete solution: Level 3 plus protection functions with circuit breakers on incoming and outgoing feeders.

What automation buys you

Better supply quality — fewer and shorter outages, improved voltage quality across the network.

Operational efficiency — enhanced network stability and better tools for operators and field crews.

Fewer site visits — less travel to hard-to-access locations; remote commands and monitoring instead.

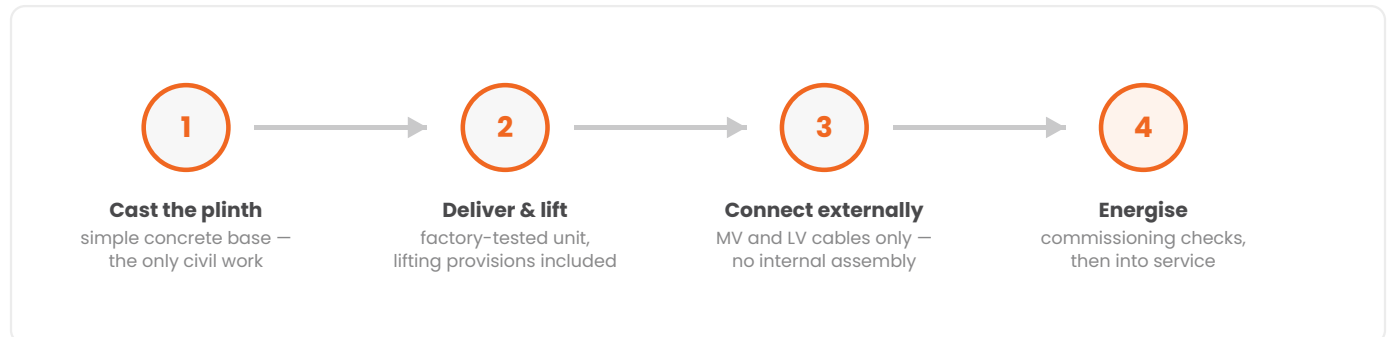
SCADA · RTU

Communication equipment and interfaces connect PCSS to any smart system through standard protocols — local and remote monitoring and commands available, and metering devices fitted per project needs.

14 · INSTALLATION

INSTALLATION & SITE WORK

PCSS is a factory solution, ready for installation with no complicated civil engineering: only a cast concrete foundation plinth needs to be ready at site.



- **Factory-tested and delivered direct** to the installation site — enclosure and equipment as one verified unit
- **Lifting provisions equipped** with the CSS; the unit can be lifted with the transformer installed
- **Only external connections** are made at site — dramatically reduced installation time
- **Lightweight, robust platform** — easy transport over long distances and less lifting effort
- **Minimal material handling** — the answer for rural sites where every site-hour counts



Delivered complete — doors locked, tested, ready to connect

Remote-ready: connect the PCSS to SCADA / RTU and site visits become the exception, not the routine.

QUALITY & TESTING

Type verification — IEC 62271-202

- **Mandatory type tests** — always performed on the design
- **Mandatory where applicable** — performed where local or national regulation requires
- **Optional type tests** — agreed between manufacturer and user for project-specific proof
- Internal arc classification passed on the standard design
- Every installed component — RMU, transformer, LV board — individually type-tested to its own product standard first

Routine tests — every unit

- Visual inspection, dimensional and build verification
- Voltage tests on main and auxiliary circuits
- Mechanical operation and interlock verification of installed switchgear
- Wiring, indication and accessory checks — documented FAT open to client witnessing



Every PCSS leaves the factory tested as a complete substation

Documentation with every delivery

Routine test certificates, as-built drawings, single-line diagram, operating manual and component type-test evidence are supplied with each PCSS.

EEHC alignment. Equipment selections follow the approved material lists of the Egyptian Electricity Holding Company throughout.

CONFIGURATION GUIDE

Four choices define your PCSS. Send us the single-line diagram — or just these answers — and Powerline engineering returns a complete, priced configuration.

1

MV switchgear

- PSEC — SF6-switch RMU
- PRAL — air-insulated RMU
- Aegis Plus — GIS RMU
- Scheme: N / F / metering ways
- 12 kV or 24 kV

2

Transformer

- Oil or dry type
- Rating up to 1,600 kVA std.
- Up to 2,500 kVA special
- Preferred manufacturer

3

LV switchboard

- Incoming ACB rating
- Outgoing feeders: number & ratings
- MCCB or switch-fuse protection
- Metering & surge protection

4

Options

- Smart grid level L1 – L4
- Urban (internal) / rural (external) service
- Stone-paint coastal finish
- Door gaskets, special IP

From enquiry to energisation

Powerline engineers the layout, manufactures and routine-tests the complete substation, delivers it to site and supports commissioning — one partner, one responsibility, one tested unit.

1

POINT OF
RESPONSIBILITY

FAT

WITNESSED BEFORE
DISPATCH



Performance excellence. Execution precision. Sustainable solutions.

From engineering and manufacture to testing, delivery and lifetime support — Powerline is your integrated partner for MV & LV power distribution in Egypt and the region.

HEAD OFFICE

20 Ammar Ibn Yasser St.
Heliopolis, Cairo
Egypt

FACTORY 1

10th of Ramadan City
Industrial Area C3
Block 112

FACTORY 2

10th of Ramadan City
Industrial Area A5
Block 143

TELEPHONE

+20 2 2621 5022

EMAIL

info@powerline.com.eg

WEB

powerlinei.com

For technical enquiries or quotations, please get in touch with our team — we respond with engineered solutions, not just prices.

24 kV

MV RATED VOLTAGE

2,500 kVA

TRANSFORMER RATINGS

5,000 A

LV SWITCHBOARD

L1 – L4

SMART GRID LEVELS